

C-CAT SECTION B

NETWORKING

Complete Revision Guide

Topics 10 – 13 | Zero to Hero

Centralized / Decentralized / Client-Server / Cloud • OSI Model • TCP/IP Protocols • IP Addressing

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Networking Basics

Core Vocabulary

- **Network** – group of connected devices sharing information/resources.
- **Node** – any device connected to a network (PC, phone, printer).
- **Server** – powerful machine that provides services to others.
- **Client** – machine that requests/uses services from a server.
- **Protocol** – agreed set of rules for communication between devices.

Network Architectures

1. Centralized Computing

- One single powerful computer called a **Mainframe** does ALL processing and storage.
- Other connected devices are **Terminals** – just a screen + keyboard, **zero processing power** of their own.
- Terminals send keystrokes to mainframe; mainframe sends back results.
- **Advantage:** Only one expensive computer needed.
- **Disadvantage: Single Point of Failure** – if mainframe crashes, every terminal goes dead. Also prone to bottleneck.

2. Decentralized Computing

- Processing and control spread across **multiple independent nodes** – no single boss/controller.
- Each node has its **own processing power** and can operate independently.
- **Advantage:** No single point of failure – if one node fails, others continue.
- **Disadvantage:** Harder to coordinate; data consistency more difficult.

3. Client-Server Architecture

- Specific model describing **roles**: Servers provide services; Clients request them.
- Server runs continuously, listening for client requests.
- Step-by-step: Client sends Request → Server processes → Server sends Response → Client uses it.
- Client has its **own processing power** (unlike terminals in centralized).
- **Advantage:** Easy to update (change data once on server; all clients see new version).

4. Cloud Computing

- Delivering computing services (storage, processing, software) over the **internet** from remote data centers.
- Built on top of the client-server model – your device is the client; cloud data centers are the servers.
- **Advantage:** No need to buy/maintain own hardware; pay only for what you use; scales easily.

Cloud Service Models

Model	Full Name	What YOU manage	Example
IaaS	Infrastructure as a Service	OS + Applications (provider gives hardware)	AWS EC2
PaaS	Platform as a Service	Only your Application (provider gives platform + infra)	Google App Engine

SaaS	Software as a Service	Nothing (provider gives ready software)	Gmail, Google Docs
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Cloud Deployment Types

- **Public Cloud** – services over public internet, shared across multiple organizations (AWS, Azure).
- **Private Cloud** – infrastructure used exclusively by one organization, often on-premises.
- **Hybrid Cloud** – combination of public and private cloud used together.

Extra: P2P and Distributed Systems

- **Peer-to-Peer (P2P)** – Every device acts as BOTH client AND server. No fixed roles. Example: BitTorrent.
- **Distributed System** – Multiple computers work together, appearing to user as ONE single system.
- **Scalability** – Ability to handle increased load by adding resources. Cloud is highly scalable.

EXAM FOCUS

Terminals (centralized) have ZERO processing power. Clients (client-server) have THEIR OWN processing power.

Cloud is built ON TOP of the client-server model – not a separate architecture.

IaaS > PaaS > SaaS in terms of 'how much YOU manage' (IaaS = most; SaaS = nothing).

P2P = roles interchangeable; Client-Server = roles fixed (server only serves).

Single point of failure = centralized. No single point of failure = decentralized.

COMMON EXAM TRAPS

Centralized vs Client-Server: In centralized, terminals do ZERO work. In client-server, clients run their own apps (browser, email) locally.

'Better' architecture depends on use case – decentralized removes failure risk but is harder to coordinate.

TOPIC 11

OSI Layer Model

What and Why

The **OSI Model** (Open Systems Interconnection, created by ISO) is a conceptual framework dividing network communication into **7 ordered layers**. Each layer has one specific job. Together they describe how data travels from sender to receiver across any network.

Why? Before OSI, devices from different manufacturers couldn't communicate. OSI created a universal blueprint so any device following these layers can talk to any other device.

Encapsulation and De-encapsulation

- **Header** – extra control information (like an address label) added to the **FRONT** of data by each layer.
- **Encapsulation** – on the **SENDER** side, each layer (7→1) wraps data with its own header (like layers of an onion).
- **De-encapsulation** – on the **RECEIVER** side, each layer (1→7) strips off and reads its own header (unwrapping the onion).

Mnemonic (Layer 7 → Layer 1)

A ll (Application)
P eople (Presentation)
S eem (Session)
T o (Transport)
N eed (Network)
D ata (Data Link)
P rocessing (Physical)

The 7 Layers – Full Reference

Layer #	Layer Name	Job / Function	Data Unit	Key Protocols / Devices
7	Application	User-facing services; what you want to do	Data	HTTP, FTP, DNS, SMTP
6	Presentation	Format translation, Encryption/Decryption, Compression	Data	TLS/SSL, JPEG, ASCII
5	Session	Establish, manage, terminate sessions between devices	Data	NetBIOS, PPTP
4	Transport	Reliable/fast delivery; segmentation; port numbers	Segment (TCP) / Datagram (UDP)	TCP, UDP
3	Network	Routing across networks; IP addressing	Packet	IP, ICMP, Router
2	Data Link	Local network delivery; MAC addressing; error detection	Frame	Ethernet, Switch, MAC

1	Physical	Actual physical transmission (electrical/light/radio signals)	Bits	Cables, Hub, NIC
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IP Address vs MAC Address – The Critical Pair

Feature	IP Address	MAC Address
OSI Layer	Layer 3 – Network	Layer 2 – Data Link
Type	Logical / Can change	Physical / Mostly permanent
Scope	Across entire internet (routing)	Within local network only
Changes during journey?	NO – stays same source to destination	YES – updated at every router hop
Analogy	Full postal address (city + street)	Fingerprint of the hardware

Devices by Layer

- **Hub** – Layer 1 (Physical). Broadcasts signal to ALL connected devices. Dumb device.
- **Switch** – Layer 2 (Data Link). Uses MAC addresses to forward frames only to the correct device.
- **Router** – Layer 3 (Network). Uses IP addresses to route packets across different networks.

EXAM FOCUS

Data unit names: Data (L7-5) → Segment (L4, TCP) / Datagram (L4, UDP) → Packet (L3) → Frame (L2) → Bits (L1).

Router = Layer 3 (IP). Switch = Layer 2 (MAC). Hub = Layer 1 (Physical). Never mix these up.

IP address does NOT change hop-to-hop. MAC address DOES change at every hop.

Encapsulation = sender, top-to-bottom, adding headers. De-encapsulation = receiver, bottom-to-top, removing headers.

TCP/UDP live at Layer 4 – Transport. IP lives at Layer 3 – Network.

COMMON EXAM TRAPS

Router vs Switch: Router routes BETWEEN networks (uses IP). Switch forwards WITHIN a network (uses MAC).

The OSI model is a REFERENCE model – not actual software/hardware. TCP/IP model (4 layers) is what's actually implemented on the internet.

TCP/IP model mapping: Application+Presentation+Session = Application | Transport = Transport | Network = Internet | Data Link+Physical = Link

TOPIC 12

Common TCP/IP Protocols

Quick Protocol Reference

Protocol	Full Name	OSI Layer	Job	Port
ARP	Address Resolution Protocol	L2/L3 boundary	IP address → MAC address (local network)	No port
IP	Internet Protocol	L3 – Network	Addressing and routing of packets	No port
ICMP	Internet Control Message Protocol	L3 – Network	Error reporting and diagnostics (ping)	No port
TCP	Transmission Control Protocol	L4 – Transport	Reliable, ordered, connection-oriented delivery	Varies
UDP	User Datagram Protocol	L4 – Transport	Fast, connectionless, best-effort delivery	Varies
FTP	File Transfer Protocol	L7 – Application	Transfer files between client and server	21
DNS	Domain Name System	L7 – Application	Domain name → IP address translation	53
HTTP	HyperText Transfer Protocol	L7 – Application	Transfer web pages (browser ↔ web server)	80
HTTPS	HTTP Secure	L7 – Application	HTTP + TLS/SSL encryption	443
SMTP	Simple Mail Transfer Protocol	L7 – Application	Sending email	25

ARP – Address Resolution Protocol

ARP answers the question: *"I know a device's IP address – but what is its MAC address?"* (needed to actually deliver a frame on the local network at Layer 2).

- **Step 1:** Device A broadcasts ARP request to ALL devices: *"Who has IP X.X.X.X? Tell me your MAC."*
- **Step 2:** Only Device B (owner of that IP) replies: *"That's me – here's my MAC: XX:XX:XX"*
- **Step 3:** Device A stores this IP→MAC mapping in its **ARP Cache/Table** for future use.
- **Broadcast** = message sent to ALL devices on a local network simultaneously.
- ARP resolves the **next-hop MAC** at each router; this is why MAC changes at every hop but IP stays constant.

ICMP – Internet Control Message Protocol

- Used for **error reporting and diagnostics**, NOT for transferring user data.
- **Ping** = ICMP Echo Request sent to a target. Target replies with ICMP Echo Reply if reachable.
- Other ICMP messages: **Destination Unreachable**, **Time Exceeded** (packet discarded after too many hops).
- ICMP does **NOT use port numbers** – it is a network-level helper protocol.

TCP – Transmission Control Protocol

- **Connection-oriented**: establishes a connection via handshake BEFORE sending data.
- **Three-Way Handshake** (memorize this exactly):
 1. **SYN** – Sender: "I want to connect"
 2. **SYN-ACK** – Receiver: "Acknowledged, I also want to connect"
 3. **ACK** – Sender: "Acknowledged – connection established"
- **ACK (Acknowledgement)**: receiver confirms each segment received. Missing ACK triggers retransmission.
- **Sequence numbers**: TCP numbers each segment so receiver reassembles them in correct order.
- **Reliable + Ordered + Error-checked** – slower due to overhead.
- **Use cases**: Web browsing (HTTP), Email (SMTP), File transfer (FTP).

UDP – User Datagram Protocol

- **Connectionless**: no handshake – data sent immediately.
- No ACKs, no retransmission, no ordering guarantee.
- **Fast** because there is no overhead. Best-effort delivery.
- **Use cases**: Live video calls, Online gaming, Video/music streaming, DNS queries.

TCP vs UDP – The Most Tested Comparison

Feature	TCP	UDP
Connection type	Connection-oriented (handshake required)	Connectionless (no handshake)
Reliability	Reliable – ACKs + retransmission	Unreliable – no guarantees
Order of delivery	Guaranteed in order	No order guarantee
Speed	Slower (more overhead)	Faster (minimal overhead)
Data unit	Segment	Datagram
Use cases	HTTP, FTP, Email, File transfer	Video call, Gaming, Streaming, DNS

DNS – Domain Name System

DNS is the **phonebook of the internet**: it converts human-readable domain names (google.com) into IP addresses computers use for routing.

- **Step 1**: You type *www.google.com* in browser.
- **Step 2**: Your computer sends query to a **DNS server**: "What is the IP for google.com?"
- **Step 3**: DNS server replies with the IP address (e.g., 142.250.193.46).
- **Step 4**: Browser uses that IP to connect to Google's server.
- DNS uses **Port 53**.

HTTP / HTTPS

- **HTTP**: protocol for transferring web pages between browser (client) and web server. Runs on TCP. Port 80.
- **HTTPS**: HTTP + **TLS/SSL encryption** – data is scrambled so outsiders cannot read it. Port 443.
- Browser sends **HTTP GET request** → Server sends back **HTTP response** (webpage HTML, images, etc.).
- HTTP runs on top of **TCP** (webpages must be delivered completely and correctly).

Full URL Journey – What Happens When You Press Enter

Step-by-step sequence (www.example.com – first visit, nothing cached)

1. DNS Lookup: Browser asks DNS server for IP of www.example.com → receives IP address.
 2. ARP: Computer needs to send data OUT of local network via its router (default gateway). ARP broadcasts to find router's MAC address.
 3. TCP Three-Way Handshake: SYN → SYN-ACK → ACK with example.com's server (establishes connection).
 4. HTTP GET Request: Browser sends HTTP GET to server over the TCP connection.
 5. HTTP Response: Server sends back webpage content. Browser renders and displays it.
- KEY ORDER: DNS → ARP → TCP Handshake → HTTP Request → HTTP Response

EXAM FOCUS

ARP: IP → MAC (local network only). DNS: Domain name → IP (internet-wide). Different problems, different scales.

TCP Three-Way Handshake: SYN, SYN-ACK, ACK. This MUST be memorized in exact order.

ICMP = diagnostics/ping. No port number. NOT for user data.

Port numbers: HTTP=80, HTTPS=443, FTP=21, DNS=53, SMTP=25.

HTTP and FTP both run on top of TCP (reliability needed). DNS often uses UDP (fast, small queries).

COMMON EXAM TRAPS

ARP vs DNS: Students confuse both as 'translation' protocols. ARP: IP→MAC (Layer 2, local). DNS: Name→IP (Layer 7, internet-wide).

UDP is NOT 'worse' than TCP – it is the RIGHT choice when speed matters more than perfect delivery (gaming, video calls).

ICMP does not use port numbers. A common wrong-answer option in MCQs.

TOPIC 13

IP Addressing

IPv4 Structure

An **IPv4 address** is a **32-bit** number written as **4 octets** (groups of 8 bits) in decimal, separated by dots. Example: **192.168.1.1**

- **Octet** = 8 bits. Each octet ranges from **0 to 255** (because $2^8 = 256$ values: 0–255).
- Total IPv4 addresses possible: $2^{32} \approx$ **4.3 billion**.
- Every IP address has two portions: **Network ID** (which network) + **Host ID** (which device on that network).
- **Subnet Mask**: a number (e.g., 255.255.255.0) that shows WHERE the dividing line is between Network and Host portions. '255' = Network part; '0' = Host part.

IPv4 Address Classes

Class	First Octet Range	Network Bits	Host Bits	Used For	Default Subnet Mask
A	1 – 126	8	24	Very large networks (few nets, millions of hosts)	255.0.0.0
B	128 – 191	16	16	Medium networks (balanced)	255.255.0.0
C	192 – 223	24	8	Small networks (most homes/offices, max 254 hosts each)	255.255.255.0
D	224 – 239	—	—	Multicasting (send to a group of devices)	Reserved
E	240 – 255	—	—	Experimental/Research use only	Reserved

- **127.x.x.x is reserved for Loopback** – specifically 127.0.0.1 = the device refers to ITSELF. Used for local testing.
- Note: 127 is skipped between Class A (ends at 126) and Class B (starts at 128) because 127.x.x.x is fully reserved.

Public IP vs Private IP

Public IP: globally unique address, routable across the entire internet. Assigned by your **ISP** (Internet Service Provider – company that gives you internet, e.g., Jio, Airtel).

Private IP: used only WITHIN a local network (home, office). Not directly reachable from the internet. The SAME private IP ranges can be reused across millions of different homes without conflict.

Reserved Private IP Ranges – Memorize These

Class	Private IP Range	Common Example
A	10.0.0.0 – 10.255.255.255	10.0.0.1 (large corporate networks)
B	172.16.0.0 – 172.31.255.255	172.16.0.1
C	192.168.0.0 – 192.168.255.255	192.168.1.1 (home WiFi routers)

NAT – Network Address Translation

NAT is the process performed by your **router** that converts multiple **private IP addresses** (devices inside your home) into ONE **public IP address** when communicating with the internet.

- **Why NAT exists:** IPv4 has only ~4.3 billion addresses, but billions of devices exist worldwide. NAT lets many devices share ONE public IP, massively extending IPv4's lifespan.
- **Outgoing:** Router replaces private IP (192.168.1.5) with its own public IP → internet sees ONE public IP.
- **Incoming:** Router receives response, translates back to the correct private IP of the internal device.

Home Network Example

Laptop: 192.168.1.5 (private) ■■■■

Phone: 192.168.1.6 (private) ■■■■■■ Router (NAT) ■■ Public IP: 49.x.x.x ■■ Internet

Smart TV: 192.168.1.7 (private) ■■■■

All 3 devices share ONE public IP. The internet only sees 49.x.x.x.

IPv4 vs IPv6

Feature	IPv4	IPv6
Bit length	32 bits	128 bits
Format	4 decimal octets separated by dots (e.g., 192.168.1.1)	8 groups of 16 bits in hexadecimal, separated by colons (e.g., 2001:0db8:85a3::8a2e:0370:7334)
Total addresses	$2^{32} \approx 4.3$ billion	$2^{128} \approx 340$ undecillion (virtually unlimited)
Status	Widely used, running out	Newer, designed to replace IPv4 permanently
NAT needed?	Yes (to conserve addresses)	No (enough addresses for every device directly)

Subnetting Basics (Conceptual)

Subnetting = dividing a network's Host portion into smaller sub-networks using a custom subnet mask (not just default Class A/B/C masks). Allows organizations to create multiple smaller, organized networks from one large IP range.

- **DHCP (Dynamic Host Configuration Protocol):** Automatically assigns IP addresses to devices when they join a network. This is why you don't manually type an IP when connecting to WiFi.
- **APIPA (169.254.x.x):** If a device fails to get an IP via DHCP, it self-assigns an address in this range. Seeing 169.254.x.x = device has no network connection / DHCP failed.
- **Default Gateway:** The router's IP address on your local network that your device sends data to when the destination is OUTSIDE your local network (e.g., 192.168.1.1 for most home routers).

EXAM FOCUS

IPv4 = 32 bits = 4 octets (0–255 each). IPv6 = 128 bits, hexadecimal, 8 groups.

Class ranges by first octet: A=1–126, B=128–191, C=192–223, D=224–239, E=240–255.

127.x.x.x = Loopback (reserved). 127.0.0.1 = device refers to itself.

Private ranges: 10.x.x.x (A), 172.16–31.x.x (B), 192.168.x.x (C). Not routable on internet.

NAT solves IPv4 address exhaustion by letting multiple devices share one public IP.

DHCP assigns IPs automatically. APIPA (169.254.x.x) = no DHCP connection.

COMMON EXAM TRAPS

Two devices in DIFFERENT homes CAN have the same private IP (e.g., both 192.168.1.5) – this is NOT a conflict. Their public IPs (via NAT) are different. Private IPs are local only.

Class A ends at 126, NOT 127. First octet 127 is reserved entirely for loopback.

IPv6 does not need NAT because its address space is large enough for every device to have a unique public IP.

MASTER QUICK REFERENCE

Everything that matters – one page

OSI Layers at a Glance

#	Layer	Unit	Protocol	Device
7	Application	Data	HTTP(80), HTTPS(443), FTP(21), DNS(53), SMTP(25)	—
6	Presentation	Data	TLS/SSL, JPEG	—
5	Session	Data	NetBIOS	—
4	Transport	Segment / Datagram	TCP, UDP	—
3	Network	Packet	IP, ICMP, ARP*	Router
2	Data Link	Frame	Ethernet, MAC	Switch
1	Physical	Bits	—	Hub, Cable

Protocol One-Line Summaries

Protocol	One-line summary
ARP	IP address → MAC address (local network, broadcast)
IP	Addressing + routing of packets across networks; source/destination IP stays constant
ICMP	Diagnostics and error reporting (ping); no port numbers
TCP	Reliable, ordered, connection-oriented; SYN→SYN-ACK→ACK handshake
UDP	Fast, connectionless, best-effort; used for streaming/gaming/video calls
FTP	File transfer, runs over TCP, Port 21
DNS	Domain name → IP address (phonebook of the internet), Port 53
HTTP	Web page transfer, runs over TCP, Port 80
HTTPS	HTTP + TLS/SSL encryption, Port 443

IPv4 Class Quick-Reference

Class	First Octet	Network/Host Bits	Subnet Mask	Special Note
A	1–126	8 / 24	255.0.0.0	126 is LAST valid A; 127 = Loopback reserved
B	128–191	16 / 16	255.255.0.0	Balanced – medium networks
C	192–223	24 / 8	255.255.255.0	Most common for homes; max 254 hosts/network

D	224–239	—	—	Multicast only
E	240–255	—	—	Experimental/Reserved

Key Terms Glossary

Term	Definition
Node	Any device on a network
Protocol	Agreed rules for communication
Encapsulation	Adding headers layer-by-layer (sender, L7→L1)
De-encapsulation	Removing headers layer-by-layer (receiver, L1→L7)
MAC Address	Physical, permanent hardware address; used within local network (Layer 2)
IP Address	Logical network address; used for routing across networks (Layer 3)
Port Number	Identifies which APPLICATION on a device should receive data
Router	Layer 3 device; routes packets between different networks using IP
Switch	Layer 2 device; forwards frames within same local network using MAC
NAT	Translates private IPs to one public IP (done by router)
DHCP	Automatically assigns IP addresses to devices on a network
Default Gateway	Router's local IP; device sends all external traffic here first
Broadcast	Message sent to ALL devices on local network simultaneously (used in ARP)
Loopback	127.0.0.1 – device refers to itself; used for local testing
Ping	ICMP Echo Request to test if a device is reachable
Three-Way Handshake	TCP connection setup: SYN → SYN-ACK → ACK
ISP	Internet Service Provider – company that provides internet access (e.g., Jio, Airtel)
IaaS/PaaS/SaaS	Cloud service models; SaaS = you manage nothing; IaaS = you manage most